

Lecture-30
Mathematics 1 (15B11MA111)

CO [C105.5]

Module: Laplace Transforms

Topics to be covered

- ❖ Dirac delta and unit step functions.

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Unit Step Function or Heaviside's unit function

Definition: The unit step function $u(t - a)$ is defined as

$$u(t - a) = \begin{cases} 0 & \text{for } t < a \\ 1 & \text{for } t \geq a \end{cases}$$

where $a \geq 0$.

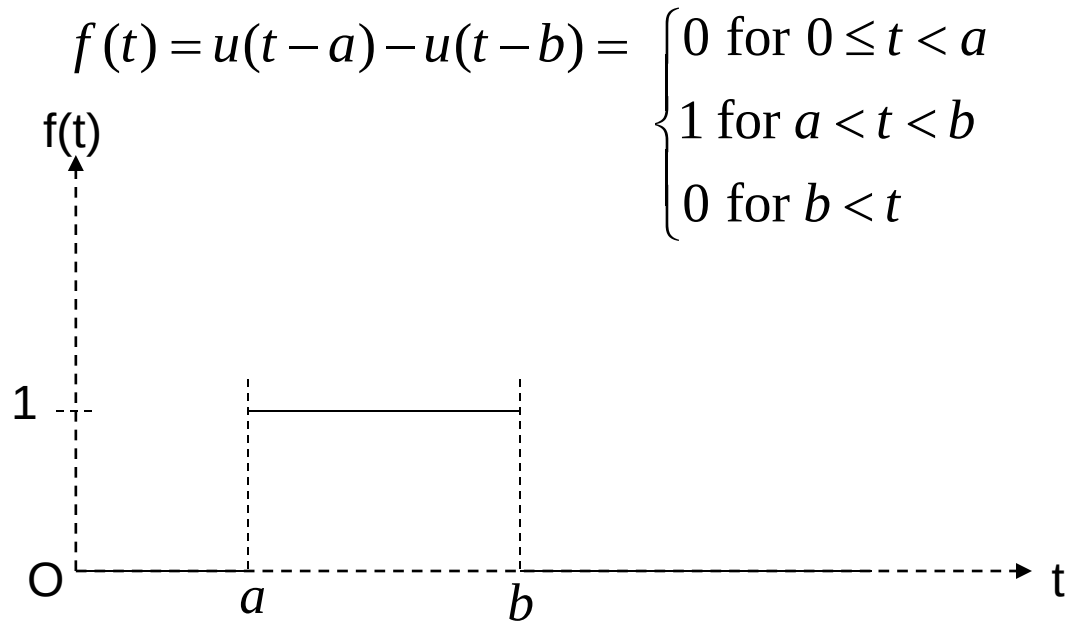
Laplace transform of Unit Step Function

$$L\{u(t - a)\} = \int_0^{\infty} e^{-st} u(t - a) dt = \int_0^a e^{-st} \cdot 0 dt + \int_a^{\infty} e^{-st} \cdot 1 dt = 0 + \left. \frac{e^{-st}}{-s} \right|_a^{\infty}$$

$$\therefore L\{u(t - a)\} = \frac{e^{-as}}{s}$$

This function is used when force or input is turned on and off for certain periods of time. When combined with a function it can produce varied results.

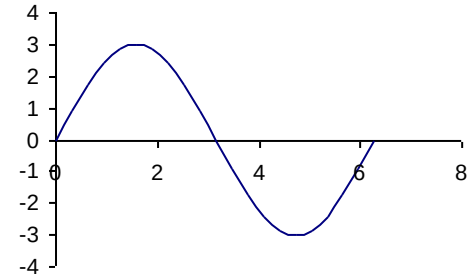
For example, the function



This function is referred to as Single Box function and represents a Single Square Wave.

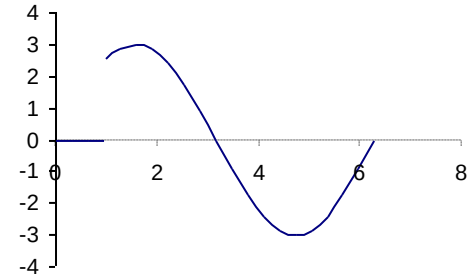
Now consider the function

$$f(t) = 3\sin t, \quad 0 \leq t \leq 2\pi$$



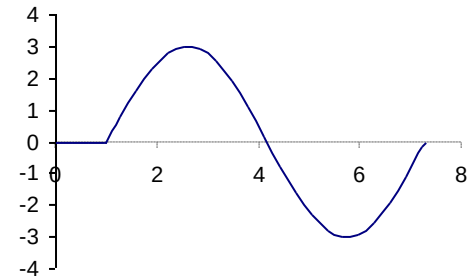
Then the function

$$f(t)u(t-1) = \begin{cases} 0 & \text{for } 0 \leq t < 1 \\ f(t) & \text{for } 1 \leq t \leq 2\pi \end{cases}$$



and the function

$$f(t-1)u(t-1) = \begin{cases} 0 & \text{for } 0 \leq t < 1 \\ f(t) & \text{for } 1 \leq t \leq 2\pi + 1 \end{cases}$$



Second Shifting Property

If $L\{f(t)\} = F(s)$, then $L\{f(t-a) \cdot u(t-a)\} = e^{-as} F(s)$

$$\begin{aligned} L\{f(t-a) \cdot u(t-a)\} &= \int_0^{\infty} f(t-a) \cdot u(t-a) e^{-st} dt \\ &= \int_a^{\infty} f(t-a) e^{-st} dt \quad \text{Put } t-a = u \\ &= \int_0^{\infty} f(u) e^{-s(u+a)} du = e^{-as} \int_0^{\infty} f(u) e^{-su} du = e^{-as} F(s) \end{aligned}$$

Applications:

1. Find the Laplace transform of $f(t) = \begin{cases} 2 & 0 < t < \pi \\ 0 & \pi < t < 2\pi \\ \sin t & t > 2\pi \end{cases}$

$$f(t) = 2u(t) - 2u(t-\pi) + u(t-2\pi) \sin t$$

$$F(s) = L\{f(t)\} = \frac{2}{s} - \frac{2}{s} e^{-\pi s} + \frac{1}{s^2 + 1} e^{-2\pi s}$$

2. Find the inverse Laplace transform of $F(s) = \frac{se^{-s/2} + \pi e^{-s}}{s^2 + \pi^2}$

$$F(s) = \frac{s}{s^2 + \pi^2} e^{-s/2} + \frac{\pi}{s^2 + \pi^2} e^{-s}$$

From second Shifting Property

$$L^{-1} \{ e^{-as} F(s) \} = f(t-a) \cdot u(t-a)$$

Since
$$L^{-1} \left\{ \frac{s}{s^2 + \pi^2} \right\} = \cos \pi t \qquad L^{-1} \left\{ \frac{\pi}{s^2 + \pi^2} \right\} = \sin \pi t$$

therefore

$$\begin{aligned} f(t) &= L^{-1} \{ F(s) \} = L^{-1} \left\{ \frac{s}{s^2 + \pi^2} e^{-s/2} \right\} + L^{-1} \left\{ \frac{\pi}{s^2 + \pi^2} e^{-s} \right\} \\ &= \cos \pi \left(t - \frac{1}{2} \right) \cdot u \left(t - \frac{1}{2} \right) + \sin \pi (t-1) \cdot u(t-1) \\ &= \sin \pi t \cdot u \left(t - \frac{1}{2} \right) - \sin \pi t \cdot u(t-1) = \left[u \left(t - \frac{1}{2} \right) - u(t-1) \right] \sin \pi t \end{aligned}$$

Unit Impulse and Dirac Delta function

Definition: Unit Impulse is defined as the integral of a large force taken over a small time interval for which it acts..

A large force acting for a small time can be defined as

$$f_k(t-a) = \begin{cases} \frac{1}{k} & a < t < a+k, k \rightarrow 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\lim_{k \rightarrow 0} f_k(t-a) = \delta(t-a)$$

The Dirac Delta function. It is not a function in true sense.

$$\delta(t-a) = \begin{cases} \infty & \text{for } t = a \\ 0 & \text{otherwise.} \end{cases} \quad \int_0^{\infty} \delta(t-a) dt = 1$$

$$f_k(t-a) = \frac{1}{k} [u(t-a) - u(t-\{a+k\})]$$

$$L\{f_k(t-a)\} = \frac{1}{ks} [e^{-as} - e^{-(a+k)s}] = \frac{1}{ks} e^{-as} [1 - e^{-ks}]$$

Taking limit as $k \rightarrow 0$, we get $L\{\delta(t-a)\} = e^{-as}$

Thank you